# EmoLearn: Adaptive Pedagogical Personalization via Multimodal Face–Voice Fusion and LLM Content Generation

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***Abstract*—Online learning platforms struggle to detect when a learner disengages, becomes confused, or frustrated. This paper presents *EmoLearn*, a research prototype that estimates engagement from facial and vocal signals, decides whether and how to intervene pedagogically, and generates adaptive learning content with a large language model (LLM) augmented by retrieval-augmented generation (RAG). The system is grounded in the DAiSEE engagement taxonomy and uses IEMOCAP for speech emotion recognition and for an aligned audiovisual fusion testbed. Component-wise evaluation shows that a learned FER-to-DAiSEE Random Forest improves accuracy from $0.193$ to $0.723$ over a heuristic mapping, and that a fine-tuned wav2vec2 speech model nearly doubles vocal accuracy relative to an acoustic heuristic. On an aligned audiovisual subset ($n$=80), weighted-vote fusion reaches $0.500$ accuracy—above either modality alone—while a pedagogical priority strategy maximizes recall of urgent states ($0.400$). We describe the end-to-end architecture, fusion trade-offs, sensitivity analyses, and the explicit limitations of the simulation prototype.**

***Index Terms*—adaptive learning, learner engagement, multimodal fusion, facial emotion recognition, speech emotion recognition, large language models, retrieval-augmented generation, DAiSEE, IEMOCAP, affective computing**

## I. Introduction

Massive Open Online Courses (MOOCs) and large-scale e-learning platforms have democratized access to education, yet they continue to exhibit high dropout rates [1]–[4]. Among the many contributing factors—time constraints, insufficient prerequisites, limited institutional recognition—a central driver is the decline of **learner engagement** [5], [6]. Engagement encompasses cognitive, behavioral, and emotional investment in the learning task. In a physical classroom, an instructor continuously reads subtle cues—averted gaze, slumped posture, puzzled expression—and adapts in real time: rephrasing, slowing down, providing an example, or re-capturing attention. That regulatory loop largely disappears in mass online settings, where learners face static content that is identical for everyone and insensitive to their momentary state.

Affective computing and educational psychology have established that emotions are not peripheral to cognition but integral to it [7], [8]. States such as boredom, confusion, and frustration influence depth of processing, persistence, and retention [9]–[11]. Detecting these states and responding appropriately is therefore a promising path toward partially restoring pedagogical regulation in online environments.

Two technological developments make this ambition increasingly credible. First, automatic engagement recognition from multimodal signals—facial expression, voice, posture, and physiological measures—has matured, supported by reference corpora such as DAiSEE for engagement [12] and IEMOCAP for conversational emotion [13], as well as facial engagement work [14]. Second, large language models (LLMs) can generate explanations, analogies, exercises, and encouragements on demand [15]–[17], especially when grounded by retrieval-augmented generation (RAG) [18].

**EmoLearn** sits at the intersection of these strands. It implements a full pipeline that (i) perceives engagement from face and voice, (ii) decides whether to intervene and which pedagogical content type to produce, and (iii) generates adaptive text via an LLM with optional RAG. The system is explicitly framed as a *simulation research prototype*: engagement signals come from annotated corpora rather than live classroom capture. This scoping choice enables controlled, reproducible evaluation of each pipeline stage.

The main contributions of this work are:

1) An end-to-end architecture linking multimodal engagement perception, rule-based pedagogical decision-making, and LLM/RAG content generation.
2) A shared fusion engine used both by a product pipeline (API + UI) and by an offline aligned audiovisual testbed for scientific validation.
3) Learned unimodal components: a Random Forest mapping FER scores to DAiSEE states, and a wav2vec2 [19] speech classifier fine-tuned for four engagement classes.
4) Empirical evidence of fusion trade-offs between global accuracy (weighted vote) and pedagogical safety (urgent-state recall), plus sensitivity analyses on fusion weights and emotion-to-engagement mapping.

The remainder of this paper is organized as follows. Sec-

tion II reviews related work. Section III presents the system architecture. Section IV details methods. Section V summarizes implementation notes. Section VI describes the experimental protocol. Section VII reports results. Section VIII discusses limitations and implications. Section IX concludes.

## II. Related Work

### A. Engagement and Affect in Learning

Discrete affective states during technology-mediated learning have been meta-analyzed by D'Mello [11]. The control–value theory of achievement emotions [8] and the dynamics of confusion and frustration [9], [10] motivate treating boredom, confusion, engagement, and frustration as pedagogically actionable targets rather than generic emotion labels alone.

### B. Automatic Engagement Recognition

Whitehill et al. [14] demonstrated automatic recognition of student engagement from facial expressions. DAiSEE [12] provides in-the-wild student videos annotated with four engagement-related states and intensity levels. Parallel advances in speech emotion recognition, including self-supervised representations such as wav2vec 2.0 [19], improve vocal affect estimation. IEMOCAP [13] remains a standard audiovisual corpus with categorical and dimensional (valence–activation–dominance) annotations.

### C. Adaptive Learning and Generative Tutors

Early adaptive systems relied on hand-authored content variants. LLMs change this economics by generating personalized explanations at scale [16]. RAG [18] reduces hallucination risk by grounding generation in a retrieved pedagogical knowledge base. Design-science perspectives [20] emphasize building and evaluating artifacts that solve identified problems—the stance adopted here.

### D. Multimodal Fusion Strategies

Late fusion (combine decisions or probability distributions after unimodal inference) remains attractive for education systems because modalities can fail independently—a face may be occluded while audio remains usable, or vice versa. Weighted averaging, confidence gating, and task-specific priority rules are common design choices. EmoLearn instantiates this design space explicitly: a default $60\%/40\%$ face/voice weighting, conflict categorization, and two operationally distinct resolvers (accuracy-oriented weighted vote versus safety-oriented pedagogical priority).

### E. Positioning of EmoLearn

Prior work often studies recognition *or* generation in isolation. EmoLearn's distinctive claim is architectural integration: a single chain from multimodal perception through pedagogical decision rules to templated LLM generation, with an explicit dual-pipeline design that separates product demonstration from aligned fusion validation. Methodologically, the project follows a design-science stance [20]: the artifact is built to address a stated problem (missing affective regulation in online learning), and evaluation focuses on whether each component and trade-off is measurable and honest rather than on maximizing a single leaderboard score.

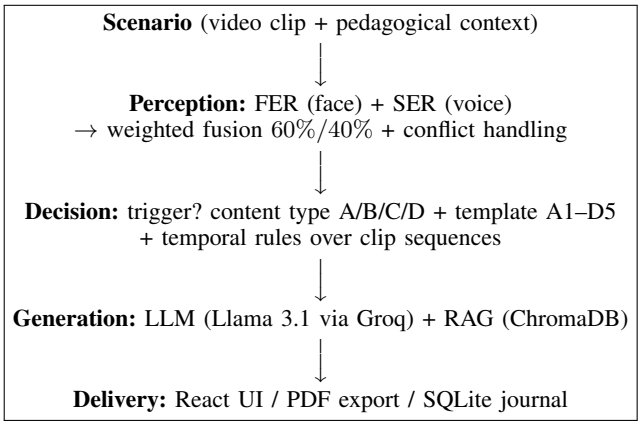



Figure 1. End-to-end EmoLearn pipeline. Both the product path and the offline testbed share the same fusion engine.

## III. System Architecture

Figure 1 summarizes the processing flow. A scenario—a video clip plus a pedagogical context—feeds emotional perception. Face and voice estimates are fused, a decision module selects intervention timing and content type, an LLM generates text (optionally with RAG), and results are delivered through a React interface with PDF export and a SQLite decision journal.

### A. Two Pipelines, One Fusion Engine

EmoLearn maintains two pipelines that must not be conflated:

- **Product pipeline.** Serves interactive adaptation: FastAPI backend, React frontend, content generation for a pedagogical scenario. Face signals may originate from DAiSEE-style engagement clips; voice may be paired by engagement state from IEMOCAP.
- **Testbed pipeline.** Offline validation of multimodal fusion on IEMOCAP segments where face and voice belong to the *same speaker* and are temporally aligned. This path is not required for the live UI demo; it exists to evaluate the fusion module under scientifically defensible pairing.

Both pipelines call the same fusion implementation. What is measured on the testbed is therefore the same decision logic used in the product path.

### B. Role of Corpora

DAiSEE is the project's conceptual backbone: all pedagogical decisions are expressed in its four-state space (Boredom, Confusion, Engagement, Frustration) with intensity levels 0–3. IEMOCAP supplies the vocal modality and the aligned audiovisual validation subset. Mapping from IEMOCAP emotion categories to DAiSEE states is a documented theoretical hypothesis (e.g., anger $\rightarrow$ Frustration), not an empirically settled equivalence—a point revisited in the sensitivity analysis.

### C. Technology Stack

The backend uses Python 3.11, FastAPI, and SQLite. Machine-learning components rely on OpenCV, librosa, scikit-learn [21], PyTorch, and Hugging Face Transformers (Vision Transformer for FER; wav2vec2 for SER). The vector store is ChromaDB with multilingual sentence embeddings. The frontend is React. Automated tests (pytest) cover API endpoints, fusion, SER, temporal rules, and RAG wiring.

## IV. Methods

### A. Facial Emotion Recognition (FER)

For each video clip, faces are detected with OpenCV Haar cascades (frontal and profile). Five frames are sampled across the clip duration. Each frame is classified by a pretrained Vision Transformer (`dima806/facial_emotions_image_detection`) into seven emotion classes (angry, disgust, fear, happy, neutral, sad, surprise). Frame-level probability vectors are aggregated by multi-frame voting (mean over valid detections).

Mapping seven FER scores to four DAiSEE states is non-trivial. We compare:

- **Heuristic mapping:** a fixed correspondence table from emotion masses to engagement states;
- **Learned mapping:** a Random Forest trained on FER score vectors labeled with DAiSEE states, with hyperparameters (`n_estimators`, `max_depth`) selected by GridSearchCV and stratified cross-validation.

The learned classifier is retained for production based on the evaluation in Section VII.

### B. Speech Emotion Recognition (SER)

Two SER methods are implemented.

*a) Heuristic acoustic baseline.:* Librosa extracts RMS energy, zero-crossing rate, spectral centroid, and pitch (`pyin`). Threshold rules estimate valence–activation–dominance (VAD) dimensions, which are then projected to DAiSEE states.

*b) Learned wav2vec2 model (default).:* We fine-tune `superb/wav2vec2-base-superb-er` [19]: the backbone is frozen and only the classification head (optionally the top layers) is trained for four DAiSEE classes. Training uses a custom PyTorch `DataLoader` with oversampling of rare classes—especially Confusion, which is absent from the original emotion head. Real IEMOCAP waveforms provide $100\%$ audio coverage for this modality.

### C. Multimodal Fusion

Let $p_f$ and $p_v$ denote face and voice probability distributions over the four DAiSEE states, and let $w_f{+}w_v{=}1$ be fusion weights. The default linear combination is

$$p = w_f\, p_f + w_v\, p_v, \qquad w_f{=}0.6,\ w_v{=}0.4. \tag{1}$$

Predicted state is $\hat{y} = \arg\max p$ for the weighted-vote variant.

Conflict detection compares unimodal predicted states and confidence levels, assigning agreement categories from perfect/partial agreement to minor/moderate/major conflict. Two configurable resolution strategies are available:

- **Pedagogical priority:** select the more urgent state according to Frustration $>$ Confusion $>$ Boredom $>$ Engagement;
- **Max confidence:** select the modality with higher confidence;
- **Weighted vote:** use (1) over full distributions (generally best for global accuracy).

### D. Temporal Triggering Rules

Adaptation is not fired on every clip. A rule engine inspects consecutive-clip history and triggers:

- **Immediate** intervention on a single critical clip;
- **Fast** intervention after 2–3 consecutive degraded states;
- **Deferred** intervention when a stable non-critical pattern persists over 4–5 clips.

This temporal smoothing reduces pedagogical noise from isolated false alarms while remaining responsive to sustained distress.

### E. Pedagogical Decision Engine

Given the fused state scores and intensity pattern, the engine selects a content *type* and then one of 20 prompt templates:

- **A** — explanation (templates A1–A5);
- **B** — worked example (B1–B5);
- **C** — practice exercise (C1–C5);
- **D** — encouragement / motivational support (D1–D5).

Combinations are allowed when mixed signals suggest both cognitive and affective support. Template choice depends on intensity and detected pattern (e.g., rising frustration vs. prolonged boredom).

### F. LLM Generation with RAG

The selected template is filled with pedagogical context, emotional state summary, and form constraints (length, tone, structure). Optional RAG retrieves passages from a vectorized pedagogical knowledge base (ChromaDB, HNSW index, cosine similarity) using `paraphrase-multilingual-MiniLM-L12-v2` embeddings. Generation uses Llama 3.1 8B Instant via the Groq API (temperature $0.3$) with retry/backoff on rate-limit errors [17], [18].

Figure 2 summarizes the online decision path as an ordered procedure used by the product API.

### G. Illustrative Scenario

Consider a short lesson segment on a difficult concept. Clip 1 yields mild Confusion from face and near-neutral voice; fusion remains below the immediate trigger. Clip 2 shows rising Confusion facially and low activation vocally; the temporal engine enters the "fast" regime. Clip 3 confirms Frustration at higher intensity. The decision module selects content type A (explanation) with a template emphasizing

**Online adaptation step**

1. Input clip $c$, pedagogical context $K$, history $H$.
2. $p_f \leftarrow$ FER($c$) — multi-frame ViT + Random Forest map.
3. $p_v \leftarrow$ SER($c$) — wav2vec2 (default) or heuristic.
4. $p \leftarrow$ Fuse($p_f, p_v$) with $w_f$=0.6.
5. $s \leftarrow$ ResolveConflicts($p_f, p_v, p$).
6. If ShouldTrigger($s, H$): select type/template $(T, \tau)$; optional RAG retrieve $R$; generate $y = \text{LLM}(\tau, K, s, R)$; log decision to SQLite; return $y$.
7. Else: update $H$; return no intervention.

Figure 2. Online adaptation control flow shared by the product pipeline.

stepwise reformulation, retrieves a relevant definition snippet via RAG, and asks the LLM for a short, calm re-explanation plus a checkpoint question. The React UI displays the generated text; the journal stores modalities, fused state, template id, and latency for later audit.

This walkthrough highlights three design intents: (i) avoid intervening on every noisy frame; (ii) escalate support as urgency accumulates; (iii) keep human-inspectable traces of why a particular adaptation was produced.

## V. Implementation Notes

The research prototype is organized as a modular Python backend and a React frontend. Perception modules expose probability distributions rather than hard labels whenever possible, so that fusion can operate on calibrated masses instead of brittle one-hot decisions. Configuration files control fusion weights, conflict strategy, SER backend (heuristic vs. learned), and whether RAG is enabled—facilitating ablation without code changes.

Persistence uses SQLite tables for sessions, per-clip predictions, fusion outcomes, selected templates, and generation metadata. This journal is essential both for debugging and for the thesis evaluation narrative: every reported number can be traced to a logged decision path. Automated tests (pytest) cover API contracts, fusion determinism for fixed inputs, temporal-rule edge cases, and RAG retrieval wiring.

From a systems perspective, LLM calls are the dominant latency and failure source. Temperature is kept low (0.3) to stabilize pedagogical tone; retries with exponential backoff absorb transient rate limits. Because the prototype is simulation-oriented, throughput is secondary to reproducibility: seeds and corpus indices are fixed for the reported testbed experiments.

## VI. Experimental Protocol

### A. Datasets

*a) DAiSEE.:* Student engagement videos with four states and levels 0–3. Used to train and test the FER $\rightarrow$ DAiSEE classifier and as the engagement ontology for decisions.

*b) IEMOCAP.:* Conversational emotional speech (and video) with categorical and VAD labels. Used for SER training/evaluation and for building an aligned audiovisual subset in which face and voice come from the same speaker.

*c) Mapping hypothesis.:* IEMOCAP categories are projected to DAiSEE states via a fixed table (e.g., anger $\rightarrow$ Frustration, sadness $\rightarrow$ Boredom by default). Because this mapping is theoretical, we quantify sensitivity to plausible alternative interpretations on all 10,039 IEMOCAP utterances.

### B. Evaluation Settings

- **FER $\rightarrow$ DAiSEE:** 1,200 real DAiSEE samples; held-out test $n$=300. Metrics: accuracy and macro-F1 (critical under class imbalance; Engagement $\approx 75\%$).
- **SER fine-tuning:** 600 IEMOCAP segments balanced across four states ($\sim$ 150 each); internal validation $n$=120; few epochs on CPU. In-deployment SER quality is primarily judged on the fusion testbed.
- **Fusion testbed:** $n$=80 aligned segments (20 per DAiSEE state). Metrics: accuracy and **urgent-state recall** (Confusion + Frustration), reflecting pedagogical safety.
- **Fusion weight sweep:** face weight from 0.5 to 0.9 on the same $n$=80 set.

### C. Baselines and Variants

Unimodal baselines include single-frame FER, multi-frame FER voting, heuristic SER, and learned SER. Fusion variants include pedagogical priority and weighted vote. The heuristic FER mapping serves as a weak baseline for the Random Forest.

### D. Why These Metrics

Engagement datasets are often imbalanced; accuracy can look strong while minority critical states are ignored. Macro-F1 therefore accompanies accuracy for FER mapping. For fusion, we additionally report *urgent-state recall* because a tutor that misses Confusion/Frustration is pedagogically riskier than one that occasionally over-helps an already engaged learner. This metric choice is a normative stance made explicit for reviewers: educational cost asymmetry should shape evaluation.

### E. Threats to Validity

Internal validity is supported by aligned speaker-synchronous testbed clips and fixed splits. External validity is limited: corpus affect is not identical to live classroom affect, and IEMOCAP conversational emotion differs from silent video-watching engagement in DAiSEE. Construct validity of the emotion-to-engagement map is the largest conceptual threat; we therefore quantify sensitivity rather than claiming the map is unique or optimal.

## VII. Results

### A. FER to DAiSEE Mapping

Table I shows that the learned Random Forest dramatically outperforms the heuristic mapping ($+0.530$ accuracy, $+0.123$ macro-F1). Macro-F1 remains modest, consistent with strong class imbalance: accuracy alone could reward majority-class prediction, so both metrics are reported. The learned model is selected for production.

Table I
FER → DAISEE CLASSIFIER (TEST $n$=300).

| Approach | Accuracy | Macro-F1 |
|---|---|---|
| Heuristic mapping (baseline) | 0.193 | 0.141 |
| Random Forest (GridSearchCV) | **0.723** | **0.264** |

Table II
WAV2VEC2 CLASSIFICATION-HEAD FINE-TUNING (VALIDATION $n$=120).

| Epoch | Accuracy | Macro-F1 |
|---|---|---|
| 1 | 0.275 | 0.279 |
| 2 | 0.300 | **0.309** |

Table III
ALIGNED AUDIOVISUAL FUSION TESTBED ($n$=80; 20 SEGMENTS PER STATE).

| Modality / approach | Acc. | Urgent recall |
|---|---|---|
| Face — FER 1 frame | 0.375 | 0.250 |
| Face — FER multi-frame vote | 0.412 | 0.275 |
| Voice — SER heuristic | 0.263 | 0.225 |
| Voice — SER wav2vec2 (learned) | 0.487 | 0.200 |
| Fusion — pedagogical priority | 0.338 | **0.400** |
| Fusion — weighted vote | **0.500** | 0.125 |

### B. SER Fine-Tuning Trajectory

Table II reports early training progress under constrained compute. Accuracy and macro-F1 improve from epoch 1 to epoch 2. These figures characterize optimization progress; absolute SER performance in the fused setting appears in Table III.

### C. Aligned Audiovisual Fusion

Table III compares modalities and fusion strategies on the aligned testbed. Multi-frame FER improves over single-frame FER (0.412 vs. 0.375). Learned SER reaches 0.487 accuracy—nearly double the heuristic SER (0.263, gain +0.224). Weighted-vote fusion achieves the best global accuracy (0.500), exceeding both unimodal systems. Pedagogical priority yields lower accuracy (0.338) but the highest urgent-state recall (0.400), illustrating a deliberate accuracy–safety trade-off.

### D. Sensitivity to Fusion Weights

Table IV varies the face weight. Accuracy peaks at $0.7/0.3$ (0.338 under the configuration used for this sweep), while urgent recall peaks at the default $0.6/0.4$ (0.425). We retain $0.6/0.4$ because urgent-state detection is the pedagogical priority of the system.

### E. Sensitivity of IEMOCAP → DAiSEE Mapping

Table V measures how alternative plausible remappings change labels across the full IEMOCAP corpus. Reinterpreting neutral as Engagement instead of Boredom alters 17.0% of states; remapping sadness to Frustration instead of Boredom alters 10.8%. Surprise and disgust reinterpretations have minor or null impact. These results localize where expert validation of the mapping would matter most.

Table IV
FUSION WEIGHT SENSITIVITY (ALIGNED SET, $n$=80).

| Face / voice weights | Acc. | Urgent recall |
|---|---|---|
| 0.5 / 0.5 | 0.287 | 0.375 |
| **0.6 / 0.4** (default) | 0.300 | **0.425** |
| 0.7 / 0.3 | **0.338** | 0.400 |
| 0.8 / 0.2 | 0.325 | 0.350 |
| 0.9 / 0.1 | 0.325 | 0.350 |

Table V
MAPPING SENSITIVITY ON FULL IEMOCAP (10,039 UTTERANCES).

| Alternative mapping | % states changed |
|---|---|
| neutral → Engagement (vs. Boredom) | 17.0% |
| sadness → Frustration (vs. Boredom) | 10.8% |
| surprise → Engagement (vs. Confusion) | 1.1% |
| disgust → Boredom (vs. Frustration) | 0.0% |

### F. Summary of Empirical Findings

1) Learned unimodal components substantially beat heuristics for both face mapping and speech classification.
2) Multimodal weighted vote can outperform the best single modality on aligned data.
3) Pedagogical priority and weighted vote optimize different objectives; deployment should choose based on whether false misses of confusion/frustration are more costly than overall accuracy.
4) Mapping assumptions are consequential and should be treated as first-class uncertainty, not as invisible preprocessing.

## VIII. DISCUSSION AND LIMITATIONS

EmoLearn deliberately exposes tensions that shorter "demo papers" often hide.

*a) Fusion versus unimodal performance.:* On the aligned testbed, weighted vote improves accuracy over unimodal baselines, resolving the question of whether fusion helps—at least under synchronized pairing. The gain is real but modest in absolute terms (0.500), consistent with a constrained $n$=80 evaluation set and the difficulty of four-way engagement classification.

*b) Emotion-to-engagement mapping.:* Projecting conversational emotions onto learning engagement remains a hypothesis. The sensitivity study shows that neutral and sadness interpretations can shift a large fraction of labels. Expert pedagogical review should prioritize these two categories.

*c) Adaptive system versus simulation.:* The prototype does not perform live webcam/microphone capture in a real MOOC. Claims are therefore about pipeline feasibility and component quality under corpus conditions, not about field efficacy.

Table VI
ASSUMED TENSIONS AND CURRENT STATUS.

| Tension | Status |
|---|---|
| Fusion vs. unimodal | Measured compromise (acc. vs. urgent recall) |
| Emotion-to-engagement map | Hypothesis + sensitivity analysis |
| Adaptive vs. simulation | Scoped as prototype (no live capture) |
| Heterogeneous face/voice pairing | Documented; mitigated via aligned subset |
| 20 templates | Argued design; expert protocol pending |
| LLM content quality | Form measurable; substance needs humans |
| Breadth vs. depth | End-to-end architecture + honest evaluation |

*d) Heterogeneous pairing in the product path.:* When the product pipeline pairs a DAiSEE face clip with an IEMOCAP voice sample by state rather than by speaker identity, modality synchrony is weaker. The aligned testbed mitigates this for fusion science; the limitation remains for the interactive demo.

*e) Templates and LLM content quality.:* The twenty templates are an argued design proposal with an evaluation protocol ready for expert review, but they are not yet validated by instructional designers. LLM output quality can be measured for form (structure, length, tone); substantive pedagogical correctness still requires human validation.

*f) Breadth versus depth.:* The central contribution is an honest end-to-end architecture with measured trade-offs, rather than a single state-of-the-art unimodal score. Depth was invested in fusion evaluation and sensitivity analyses; live deployment and learning-outcome studies are left to future work.

Table VI consolidates these points.

### A. Ethical and Practical Considerations

Affective sensing in education raises privacy and consent issues. Even as a corpus-based prototype, EmoLearn is designed so that decision logs remain inspectable (SQLite journal), supporting auditability. Any future live deployment would require explicit consent, data minimization, on-device processing where feasible, and the option for learners to disable sensing while retaining non-affective tutoring. Transparency also matters: learners and instructors should be able to understand, at least at a high level, why an adaptation was triggered (state, modality agreement, template family).

### B. Implications for Adaptive Learning Systems

The results suggest that multimodal fusion should be configured for the *loss function of teaching*, not only for classification accuracy. A tutor that slightly over-triggers on confusion may be preferable to one that optimizes accuracy while missing frustration. Separating product demonstration from aligned scientific validation is also a methodological recommendation for similar multimodal education systems.

More broadly, generative tutors without perception risk producing unsolicited or poorly timed help; perception modules without generation risk stopping at dashboards that instructors never have time to act upon. EmoLearn argues for closing the loop, while keeping humans responsible for validating pedagogical substance.

### C. Reproducibility

Reported fusion numbers correspond to a fixed aligned subset construction procedure and shared fusion code path. Unimodal training recipes (GridSearchCV for the Random Forest; frozen-backbone wav2vec2 head training with oversampling) are documented in the project repository accompanying the thesis. Because DAiSEE and IEMOCAP require academic agreements, raw media are not redistributed; scripts assume the standard corpus layouts under a local `data/` directory.

## IX. CONCLUSION AND FUTURE WORK

This paper introduced EmoLearn, a research prototype for adaptive pedagogical personalization driven by multimodal face–voice engagement estimation and LLM/RAG content generation. The system combines a product-facing pipeline with an offline aligned audiovisual testbed that share one fusion engine, enabling both demonstration and controlled measurement.

On component benchmarks, learned FER and SER models substantially outperform heuristics: FER-to-DAiSEE accuracy rises from $0.193$ to $0.723$, and learned SER nearly doubles vocal accuracy relative to an acoustic baseline. On the aligned testbed ($n$=80), weighted-vote fusion reaches $0.500$ accuracy—above either modality alone—while pedagogical priority maximizes urgent-state recall ($0.400$). Weight and mapping sensitivity analyses show that default parameters are not arbitrary: $0.6/0.4$ favors urgent recall, and neutral/sadness remappings can alter up to $17\%$ of projected states.

Future work includes: (i) expert validation of the twenty pedagogical templates; (ii) synchronized live multimodal capture in authentic learning settings with informed consent; (iii) rubric-based human evaluation of generated content substance; (iv) controlled studies linking adaptation to learning outcomes (comprehension, persistence, dropout proxies), beyond classification metrics; (v) stronger calibration and uncertainty-aware triggering to further reduce harmful false negatives on critical states; and (vi) fairness analyses across demographic and recording-condition subgroups once live data collection is ethically approved.

In summary, EmoLearn demonstrates a practical path toward affect-aware generative tutoring, provided that architectural integration is paired with explicit trade-offs, inspectable decisions, and an honest account of what a simulation prototype can—and cannot—claim.

## ACKNOWLEDGMENT

The authors thank the thesis supervisor and the providers of the DAiSEE and IEMOCAP corpora under academic agreements.

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